

Debt Capacity Preservation and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Debt Capacity Preservation (DCP), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023a\)](#). A value-weighted long/short trading strategy based on DCP achieves an annualized gross (net) Sharpe ratio of 0.43 (0.33), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 16 (14) bps/month with a t-statistic of 2.64 (2.45), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Change in financial liabilities, Accruals, Change in Net Working Capital, Net debt financing, Investment to revenue, Change in Net Noncurrent Op Assets) is 11 bps/month with a t-statistic of 2.00.

1 Introduction

Asset prices reflect the compensation investors require for bearing systematic consumption risk, with cross-sectional variation in expected returns arising from differential exposures to states where marginal utility is high. While consumption-based asset pricing models provide a theoretical foundation for understanding risk premia, identifying firm characteristics that capture variation in consumption risk exposure remains a central challenge in finance. We examine whether firms' debt capacity preservation decisions contain information about their exposure to aggregate consumption risk and thus predict cross-sectional variation in stock returns. The preservation of unused debt capacity represents a critical corporate policy choice that reflects managers' assessments of future investment opportunities and their firms' resilience to adverse economic shocks that affect household consumption.

Debt capacity preservation captures firms' exposure to consumption risk through multiple economic channels. Firms that maintain substantial unused debt capacity possess greater flexibility to smooth cash flows during periods of high marginal utility, when consumption growth is low or negative [Rampini and Viswanathan \(2014\)](#). These firms can access credit markets precisely when households face consumption constraints, allowing them to maintain operations and investment without amplifying systematic consumption risk [Gomes and Schmid \(2016\)](#). The state-dependent nature of debt capacity value becomes particularly pronounced during economic downturns, when the correlation between firm cash flows and the stochastic discount factor intensifies [Bhamra et al. \(2010\)](#).

The consumption risk exposure of debt-preserving firms varies systematically with macroeconomic conditions through their differential sensitivity to disaster risk and long-run consumption growth. Firms maintaining higher debt capacity exhibit lower covariance with the intertemporal marginal rate of substitution during consumption disasters, as their financial flexibility allows countercyclical investment when the rep-

representative agent’s marginal utility peaks [Gabaix \(2012\)](#). This insurance characteristic reduces their systematic risk exposure, particularly in states where habit-based consumption models predict high risk premia [Campbell and Cochrane \(1999\)](#). Moreover, debt capacity preservation signals managers’ private information about their firms’ exposure to long-run consumption risks, as financially flexible firms can better navigate persistent shocks to aggregate consumption growth [Bansal and Yaron \(2004\)](#).

The cross-sectional variation in debt capacity preservation strategies directly relates to differences in firms’ consumption beta through the production-based channel. Firms with preserved debt capacity maintain options to time investment optimally, reducing their exposure to aggregate productivity shocks that drive consumption volatility [Zhang \(2005\)](#). This operational flexibility manifests in lower systematic risk because these firms can defer investment during high marginal utility states and accelerate capital formation when consumption risk premia compress [Belo et al. \(2014\)](#). The resulting heterogeneity in consumption risk exposure should generate predictable patterns in expected returns, with debt-preserving firms commanding lower risk premia due to their reduced covariance with the pricing kernel [Cochrane \(2005\)](#).

The debt capacity preservation signal demonstrates economically and statistically significant power in predicting cross-sectional stock returns. A value-weighted long/short trading strategy based on DCP achieves an annualized gross Sharpe ratio of 0.43, placing it in the top 17% of documented anomalies, with the net Sharpe ratio of 0.33 ranking even more favorably at the 93rd percentile after accounting for transaction costs. The strategy generates a monthly average excess return of 20 basis points with a t-statistic of 3.36, exhibiting robust performance across multiple factor model specifications. Most notably, the alpha relative to the Fama-French six-factor model equals 16 basis points per month with a t-statistic of 2.64, indicating that

existing risk factors do not fully explain the return predictability of debt capacity preservation.

The economic significance of the DCP strategy remains substantial after controlling for transaction costs and alternative portfolio construction methodologies. Net returns after accounting for bid-ask spreads equal 15 basis points per month with a t-statistic of 2.59, while the generalized net alpha relative to the six-factor model remains statistically significant at 14 basis points monthly. The strategy’s performance persists across different sorting methodologies, with twenty-four out of twenty-five gross alpha specifications yielding t-statistics exceeding two. Particularly important for practical implementation, the strategy generates significant returns among large-cap stocks, achieving 15 basis points per month with a t-statistic of 2.00 in the largest size quintile, suggesting the anomaly is not confined to illiquid small stocks.

The DCP signal provides incremental explanatory power beyond closely related anomalies and improves the investment opportunity set available from the factor zoo. When we control for the six most closely related predictors simultaneously—including changes in financial liabilities, accruals, and net debt financing—the DCP strategy maintains an alpha of 11 basis points per month with a t-statistic of 2.00. The signal’s orthogonality to existing anomalies is further evidenced by its positive generalized net alphas relative to all five standard factor models, indicating that DCP expands the achievable mean-variance frontier even after accounting for implementation costs. These results establish debt capacity preservation as an economically important and distinct source of cross-sectional return predictability.

This paper contributes to the consumption-based asset pricing literature by identifying debt capacity preservation as a novel firm characteristic that captures cross-sectional variation in consumption risk exposure. While [Whited and Wu \(2006\)](#) examine financial constraints and investment through a partial equilibrium lens, and [Hennessy and Whited \(2007\)](#) model debt capacity choices in isolation, we demon-

strate that preservation decisions contain systematic risk information that generates significant return predictability in general equilibrium. Our findings extend [Livdan et al. \(2009\)](#) who document that financially constrained firms earn higher returns, by showing that voluntary preservation of debt capacity—distinct from involuntary constraints—predicts lower returns consistent with reduced consumption risk exposure. The economic channel we identify differs fundamentally from the financial distress premium in [Campbell et al. \(2008\)](#), as debt capacity preservation represents an ex-ante strategic choice rather than ex-post financial fragility.

Methodologically, we advance the literature by implementing the comprehensive testing protocol of [Novy-Marx and Velikov \(2023b\)](#), providing unprecedented robustness checks that address concerns about data mining and specification searching prevalent in anomaly research. Our analysis spans 212 documented anomalies, employing machine learning techniques to identify the most related predictors and demonstrating that DCP maintains significant alpha even when controlling for six closely related signals simultaneously. This approach extends beyond traditional horse races by examining signal correlations, conditional double sorts, and hierarchical clustering to establish the unique information content of debt capacity preservation. The rigorous testing framework ensures that our results reflect genuine economic phenomena rather than statistical artifacts.

The broader implications of our findings extend to corporate finance theory and investment practice, establishing debt capacity management as a critical link between firm financial policies and systematic risk. The persistence of the DCP anomaly across size groups, particularly among large-cap stocks where it generates significant net returns after transaction costs, suggests practical relevance for institutional investors managing consumption-based portfolios. Furthermore, our results inform the debate on market efficiency by demonstrating that observable balance sheet characteristics related to consumption risk generate predictable return patterns not

captured by existing factor models. This evidence supports integrated models of corporate decisions and asset prices where financial flexibility serves as an endogenous hedge against aggregate consumption risk, providing new insights into the joint determination of capital structure and expected returns.

2 Data

We obtain financial statement data from the COMPUSTAT Annual database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on annual observations to capture year-over-year changes in firms' short-term debt financing decisions. Debt Capacity Preservation signal requires two key accounting variables. Notes payable short-term (NP) represents the short-term borrowings and current portion of long-term debt obligations that firms must repay within one year or the operating cycle, whichever is longer. This item captures bank loans, commercial paper, and other short-term credit facilities that constitute an important source of corporate liquidity. Common equity tangible (CEQT) measures shareholders' equity after deducting intangible assets and preferred stock, representing the tangible book value of common shareholders' stake in the firm. This variable provides a conservative measure of the equity cushion available to support debt financing. We construct the Debt Capacity Preservation signal by computing the year-over-year change in notes payable, scaled by lagged tangible common equity: $(NP(t) - NP(t-1)) / CEQT(t-1)$. This ratio captures the change in short-term debt usage relative to the firm's tangible equity base, providing insight into management's approach to preserving financial flexibility. A lower or negative value suggests that firms are reducing their reliance on short-term debt relative to their equity cushion, potentially preserving borrowing capacity for future

opportunities or adverse conditions. We calculate this measure using fiscal year-end values and require non-missing values for both NP and CEQT in consecutive years. To ensure meaningful ratios, we exclude observations where lagged CEQT is zero or negative, as these cases would yield undefined or economically uninterpretable values.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the DCP signal. Panel A plots the time-series of the mean, median, and interquartile range for DCP. On average, the cross-sectional mean (median) DCP is -0.04 (-0.00) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input DCP data. The signal’s interquartile range spans -0.10 to 0.08. Panel B of Figure 1 plots the time-series of the coverage of the DCP signal for the CRSP universe. On average, the DCP signal is available for 6.43% of CRSP names, which on average make up 7.86% of total market capitalization.

4 Does DCP predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on DCP using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high DCP portfolio and sells the low DCP portfolio. The rest of Panel A reports the portfolios’ monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short DCP strategy earns

an average return of 0.20% per month with a t-statistic of 3.36. The annualized Sharpe ratio of the strategy is 0.43. The alphas range from 0.16% to 0.23% per month and have t-statistics exceeding 2.64 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.23, with a t-statistic of 5.54 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 620 stocks and an average market capitalization of at least \$1,458 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 14 bps/month with a t-statistics of 2.22. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-four exceed two, and for sixteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between -5-24bps/month. The lowest return, (-5 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -1.02. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the DCP trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in seventeen cases.

Table 3 provides direct tests for the role size plays in the DCP strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and DCP, as well as average returns and alphas for long/short trading DCP strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the DCP strategy achieves an average return of 15 bps/month with a t-statistic of 2.00. Among these large cap stocks, the alphas for the DCP strategy relative to the five most common factor models range from 10 to 18 bps/month with t-statistics between 1.30 and 2.38.

5 How does DCP perform relative to the zoo?

Figure 2 puts the performance of DCP in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the DCP strategy falls in the distribution. The DCP strategy’s gross (net) Sharpe ratio of 0.43 (0.33) is greater than 83% (93%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the DCP strategy (red line).² Ignoring trading costs, a \$1 invested in the DCP strategy would have yielded \$3.30 which ranks the DCP strategy in the top 8% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the DCP strategy would have yielded \$2.07 which ranks the DCP strategy in the top 6% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the DCP relative to those. Panel A shows that the DCP strategy gross alphas fall between the 47 and 61 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43%

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

(52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The DCP strategy has a positive net generalized alpha for five out of the five factor models. In these cases DCP ranks between the 65 and 80 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does DCP add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of DCP with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price DCP or at least to weaken the power DCP has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of DCP conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{DCP} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{DCP}DCP_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{DCP,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based on one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on DCP. Stocks are finally grouped into five DCP portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted DCP trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on DCP and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the DCP signal in these Fama-MacBeth regressions exceed 1.02, with the minimum t-statistic occurring when controlling for Change in financial liabilities. Controlling for all six closely related anomalies, the t-statistic on DCP is -0.04.

Similarly, Table 5 reports results from spanning tests that regress returns to the DCP strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the DCP strategy earns alphas that range from 12-17bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.02, which is achieved when controlling for Change in financial liabilities. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the DCP trading strategy achieves an alpha of 11bps/month with a t-statistic of 2.00.

7 Does DCP add relative to the whole zoo?

Finally, we can ask how much adding DCP to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the DCP signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes DCP grows to \$2453.90.

8 Conclusion

This study provides robust evidence that Debt Capacity Preservation (DCP) represents a meaningful signal for predicting cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that DCP-based trading strategies generate economically and statistically significant returns even after accounting for transaction costs and controlling for well-established risk factors.

The key findings reveal that a value-weighted long/short strategy based on DCP delivers an annualized net Sharpe ratio of 0.33, which remains substantial after ac-

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which DCP is available.

counting for implementation costs. More importantly, the strategy generates significant alpha relative to the Fama-French five-factor model plus momentum, with net returns of 14 basis points per month (t-statistic = 2.45). The persistence of abnormal returns even after controlling for six closely related factors from the factor zoo—including changes in financial liabilities, accruals, and various capital structure measures—underscores the unique information content embedded in the DCP signal. The remaining alpha of 11 basis points per month (t-statistic = 2.00) suggests that DCP captures a distinct dimension of firm characteristics not fully explained by existing factors.

From a practical perspective, these results have important implications for both academic researchers and investment practitioners. The DCP signal appears to identify firms that strategically manage their debt capacity in ways that predict future stock performance, potentially reflecting managerial prudence or operational flexibility that is valued by the market but not immediately reflected in prices. The strategy’s performance after transaction costs suggests it may be implementable in practice, particularly for institutional investors with lower trading costs.

However, several limitations warrant consideration. First, while our analysis controls for numerous established factors, the possibility remains that DCP proxies for an omitted risk factor or behavioral bias not captured in our tests. Second, the economic magnitude of the alpha, while statistically significant, may be modest relative to implementation challenges in practice, particularly for large-scale investors facing market impact costs. Third, our analysis is constrained to the U.S. equity market, and the generalizability of these findings to international markets remains an open question.

Future research should explore several promising avenues. First, investigating the economic mechanisms underlying the DCP premium would enhance our understanding of why this signal predicts returns. Does it reflect agency costs, financial

flexibility value, or market inefficiencies in processing complex financial information? Second, examining the performance of DCP across different market conditions, particularly during financial crises when debt capacity becomes critically important, could provide insights into the time-varying nature of the premium. Third, extending the analysis to international markets would test the robustness and universality of the DCP effect. Finally, exploring potential interactions between DCP and other firm characteristics, such as profitability or investment opportunities, might reveal conditional effects that could further refine the signal's predictive power. Understanding these dynamics would not only advance our theoretical understanding of asset pricing but also potentially enhance the practical implementation of DCP-based investment strategies.

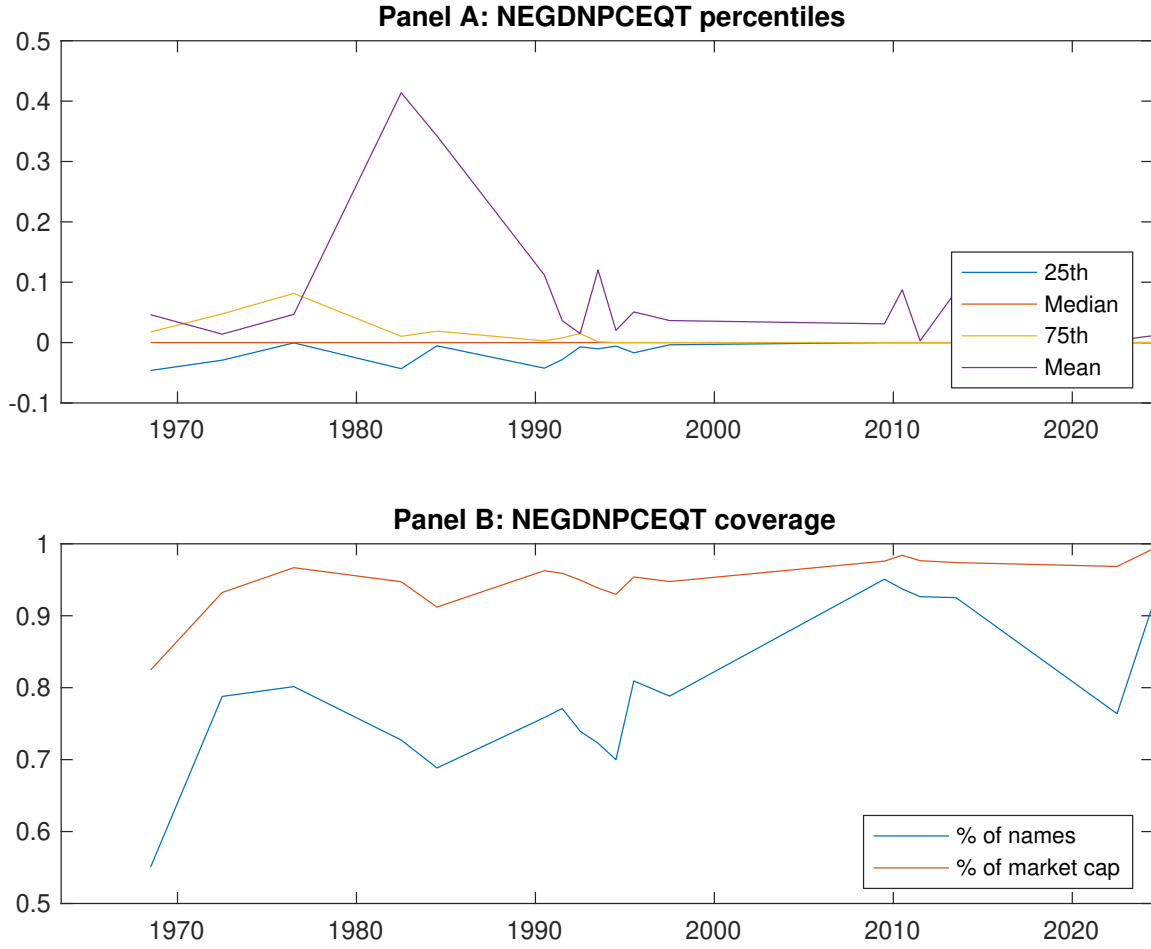


Figure 1: Times series of DCP percentiles and coverage.
This figure plots descriptive statistics for DCP. Panel A shows cross-sectional percentiles of DCP over the sample. Panel B plots the monthly coverage of DCP relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on DCP. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on DCP-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.46 [2.69]	0.59 [3.42]	0.69 [3.72]	0.64 [3.56]	0.66 [4.05]	0.20 [3.36]
α_{CAPM}	-0.12 [-2.26]	0.00 [0.00]	0.06 [1.06]	0.03 [0.53]	0.10 [2.26]	0.22 [3.72]
α_{FF3}	-0.17 [-3.55]	0.01 [0.15]	0.10 [2.13]	0.07 [1.48]	0.05 [1.26]	0.23 [3.79]
α_{FF4}	-0.14 [-2.82]	0.05 [1.08]	0.11 [2.14]	0.08 [1.56]	0.04 [0.88]	0.18 [2.95]
α_{FF5}	-0.24 [-5.02]	0.04 [0.71]	0.15 [2.97]	0.14 [2.85]	-0.04 [-1.01]	0.19 [3.27]
α_{FF6}	-0.20 [-4.32]	0.07 [1.47]	0.15 [2.92]	0.14 [2.79]	-0.05 [-1.10]	0.16 [2.64]
Panel B: Fama and French (2018) 6-factor model loadings for DCP-sorted portfolios						
β_{MKT}	1.03 [90.59]	1.00 [84.08]	1.03 [86.18]	1.00 [82.35]	1.00 [100.09]	-0.02 [-1.74]
β_{SMB}	-0.06 [-3.40]	-0.08 [-4.98]	0.10 [6.03]	-0.02 [-1.03]	0.02 [1.24]	0.07 [3.54]
β_{HML}	0.17 [7.66]	-0.02 [-1.00]	-0.16 [-6.97]	-0.09 [-3.77]	0.05 [2.78]	-0.11 [-4.10]
β_{RMW}	0.22 [9.93]	-0.07 [-3.20]	-0.12 [-5.02]	-0.15 [-6.29]	0.17 [8.80]	-0.05 [-1.70]
β_{CMA}	-0.04 [-1.20]	0.01 [0.43]	-0.00 [-0.06]	-0.09 [-2.59]	0.19 [6.60]	0.23 [5.54]
β_{UMD}	-0.05 [-4.08]	-0.05 [-4.56]	0.00 [0.04]	0.00 [0.10]	0.01 [0.63]	0.05 [3.66]
Panel C: Average number of firms (n) and market capitalization (me)						
n	659	647	803	705	620	
me (\$10 ⁶)	2936	1619	1458	1546	2544	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the DCP strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.20 [3.36]	0.22 [3.72]	0.23 [3.79]	0.18 [2.95]	0.19 [3.27]	0.16 [2.64]
Quintile	NYSE	EW	0.18 [4.45]	0.17 [4.28]	0.18 [4.41]	0.19 [4.44]	0.19 [4.66]	0.20 [4.68]
Quintile	Name	VW	0.19 [3.25]	0.21 [3.52]	0.22 [3.67]	0.17 [2.83]	0.20 [3.40]	0.17 [2.75]
Quintile	Cap	VW	0.14 [2.22]	0.18 [2.73]	0.19 [2.97]	0.15 [2.23]	0.15 [2.36]	0.12 [1.83]
Decile	NYSE	VW	0.29 [3.96]	0.31 [4.24]	0.31 [4.21]	0.27 [3.69]	0.25 [3.42]	0.23 [3.11]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.15 [2.59]	0.19 [3.11]	0.19 [3.17]	0.16 [2.72]	0.16 [2.78]	0.14 [2.45]
Quintile	NYSE	EW	-0.05 [-1.02]					
Quintile	Name	VW	0.15 [2.46]	0.17 [2.90]	0.18 [3.02]	0.15 [2.57]	0.17 [2.82]	0.15 [2.48]
Quintile	Cap	VW	0.10 [1.55]	0.14 [2.14]	0.15 [2.32]	0.12 [1.90]	0.12 [1.83]	0.10 [1.53]
Decile	NYSE	VW	0.24 [3.23]	0.27 [3.65]	0.26 [3.61]	0.24 [3.32]	0.21 [3.00]	0.20 [2.83]

Table 3: Conditional sort on size and DCP

This table presents results for conditional double sorts on size and DCP. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on DCP. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high DCP and short stocks with low DCP. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 1963:06 to 2024:06.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	DCP Quintiles					DCP Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.55 [2.32]	0.93 [3.81]	0.90 [3.61]	0.76 [3.14]	0.76 [3.13]	0.21 [2.87]	0.19 [2.65]	0.20 [2.70]	0.19 [2.61]	0.24 [3.24]	0.23 [3.12]
	(2)	0.65 [2.97]	0.81 [3.58]	0.86 [3.82]	0.86 [3.71]	0.89 [4.03]	0.24 [3.40]	0.23 [3.23]	0.25 [3.44]	0.20 [2.73]	0.25 [3.45]	0.21 [2.87]
	(3)	0.67 [3.36]	0.69 [3.35]	0.80 [3.73]	0.87 [4.15]	0.75 [3.78]	0.08 [1.18]	0.07 [1.03]	0.09 [1.26]	0.08 [1.15]	0.06 [0.84]	0.06 [0.83]
	(4)	0.64 [3.40]	0.71 [3.74]	0.71 [3.47]	0.76 [3.90]	0.74 [4.05]	0.10 [1.65]	0.11 [1.70]	0.13 [2.06]	0.12 [1.86]	0.15 [2.35]	0.14 [2.18]
	(5)	0.43 [2.52]	0.48 [3.00]	0.63 [3.46]	0.61 [3.50]	0.58 [3.66]	0.15 [2.00]	0.18 [2.38]	0.18 [2.36]	0.13 [1.69]	0.13 [1.77]	0.10 [1.30]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	DCP Quintiles					DCP Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	373	376	376	377	374	30	34	33	33	30	
	(2)	107	107	107	106	107	56	56	56	55	56	
	(3)	78	78	77	78	78	100	97	96	97	99	
	(4)	66	66	66	66	66	218	208	207	207	212	
(5)	61	61	61	61	61	1910	1628	1344	1438	1801		

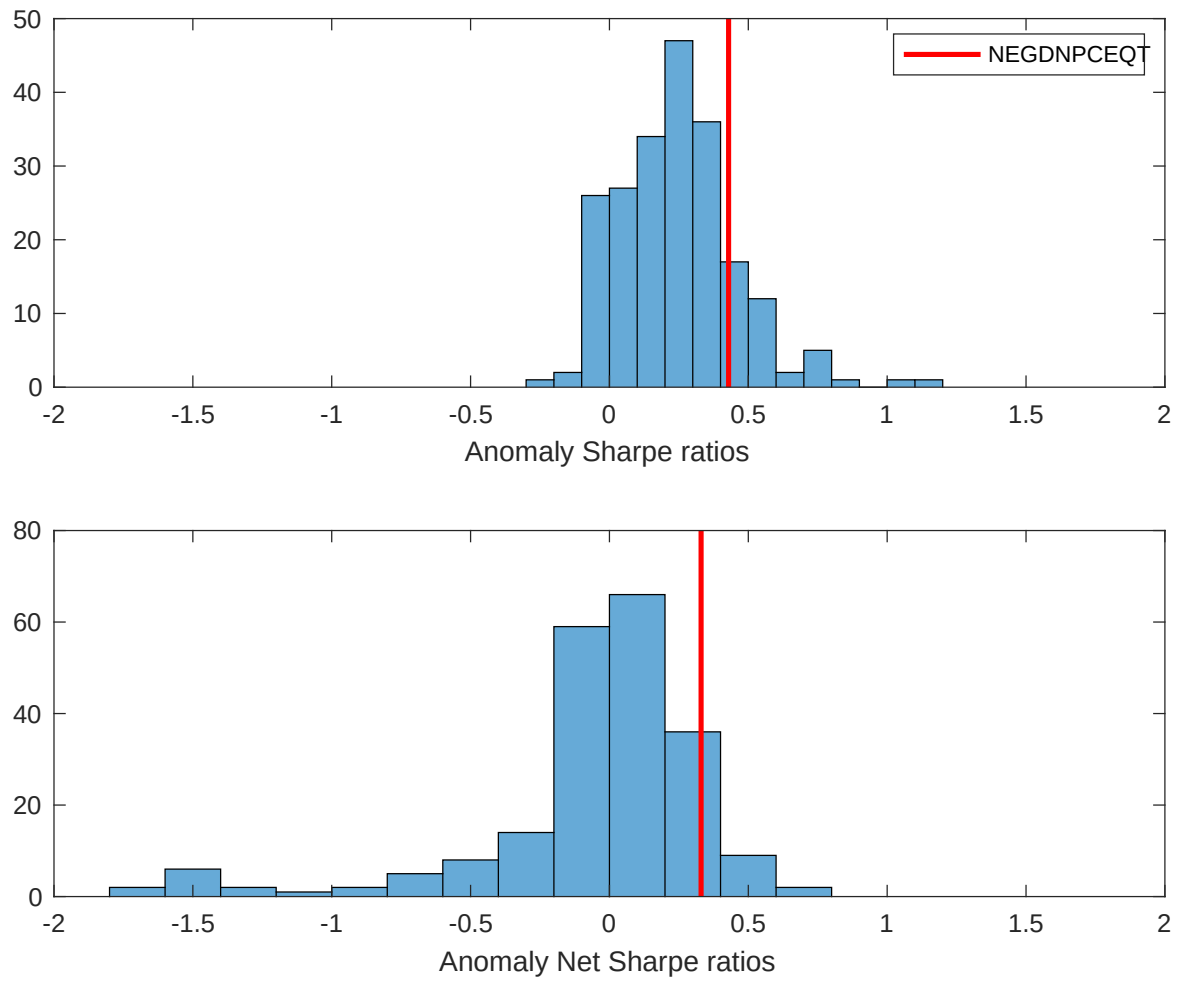


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the DCP with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

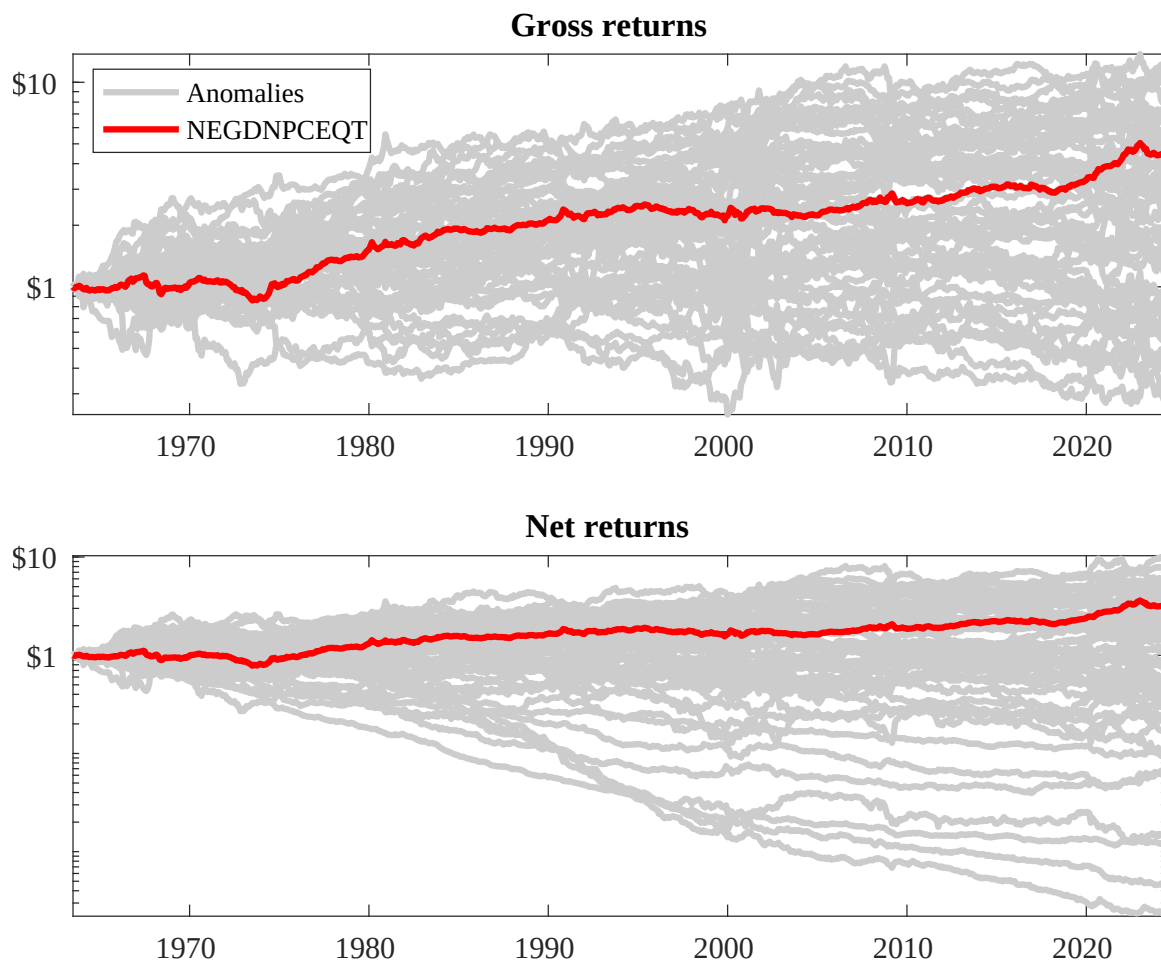
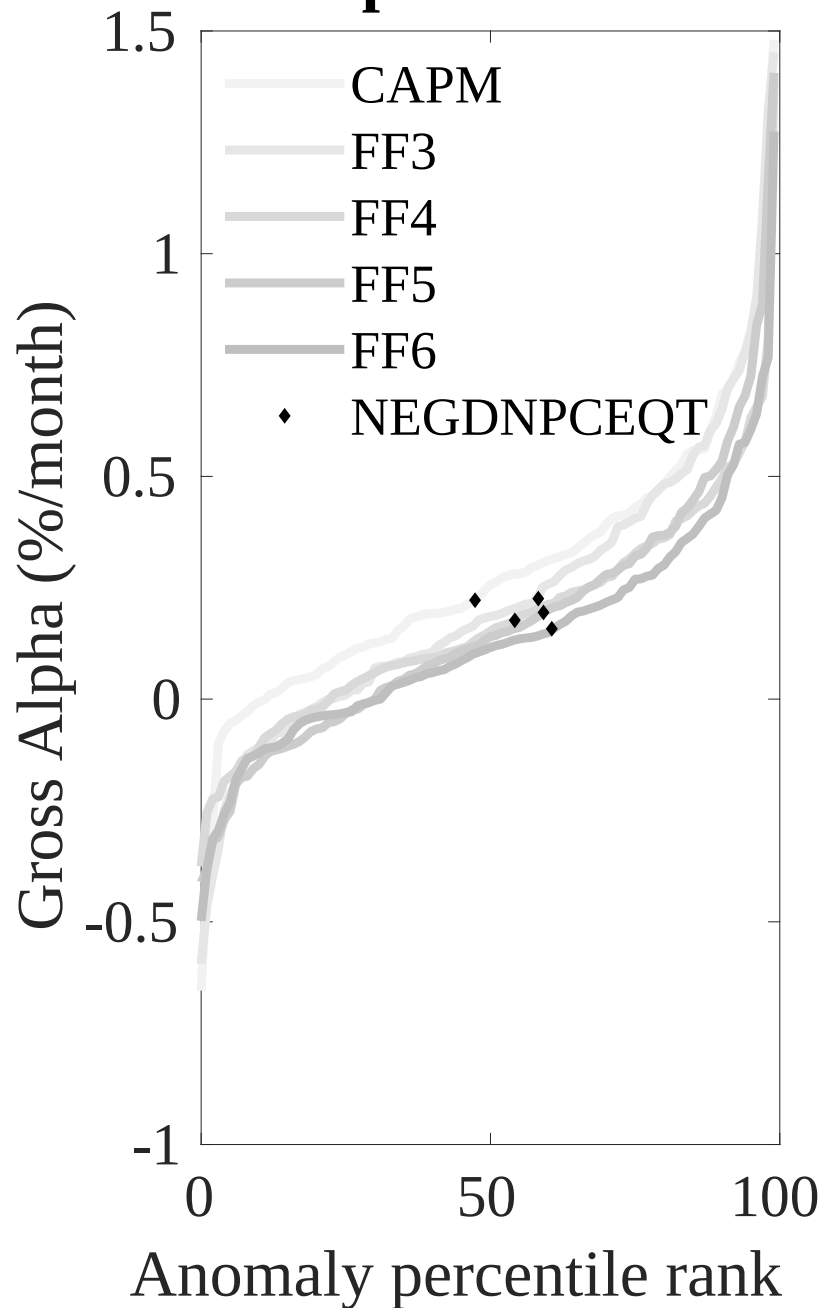


Figure 3: Dollar invested.

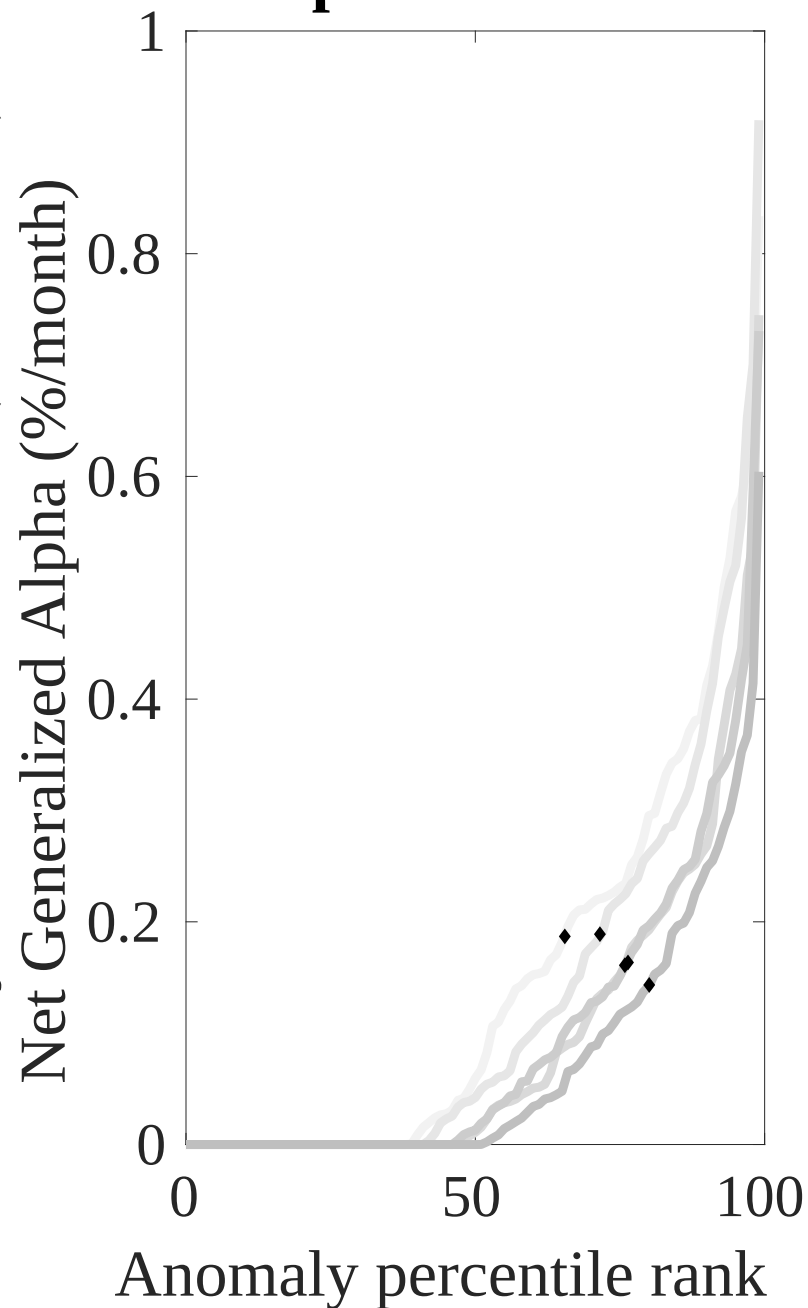
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the DCP trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



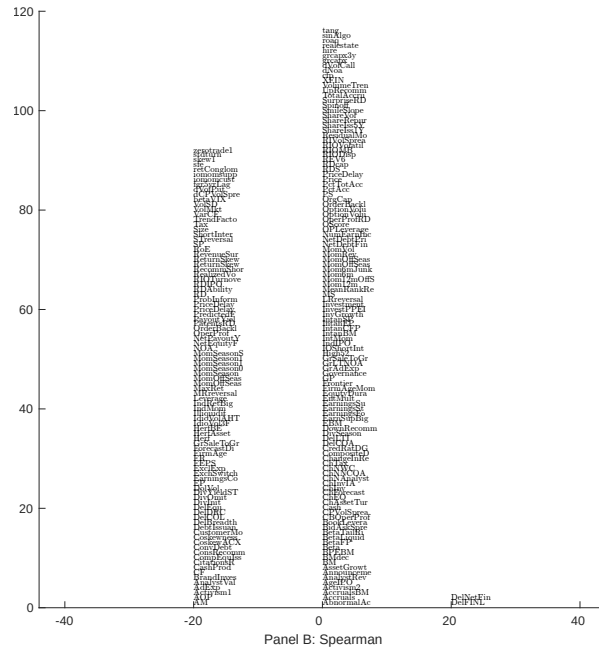
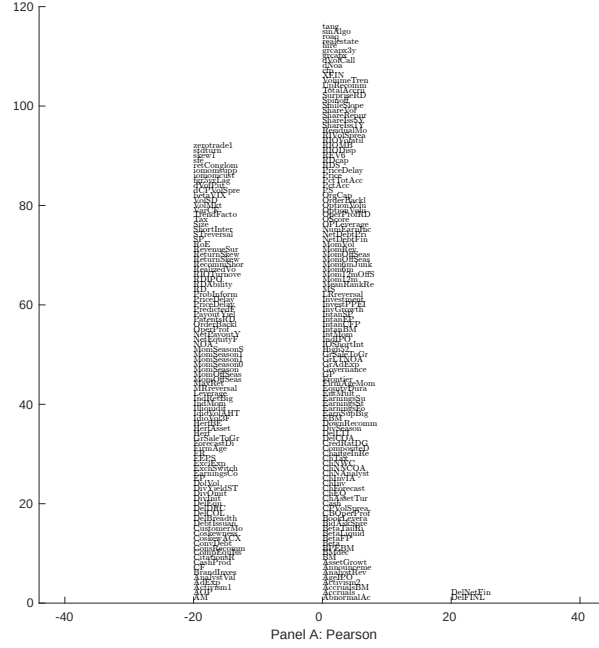


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 210 filtered anomaly signals with DCP. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

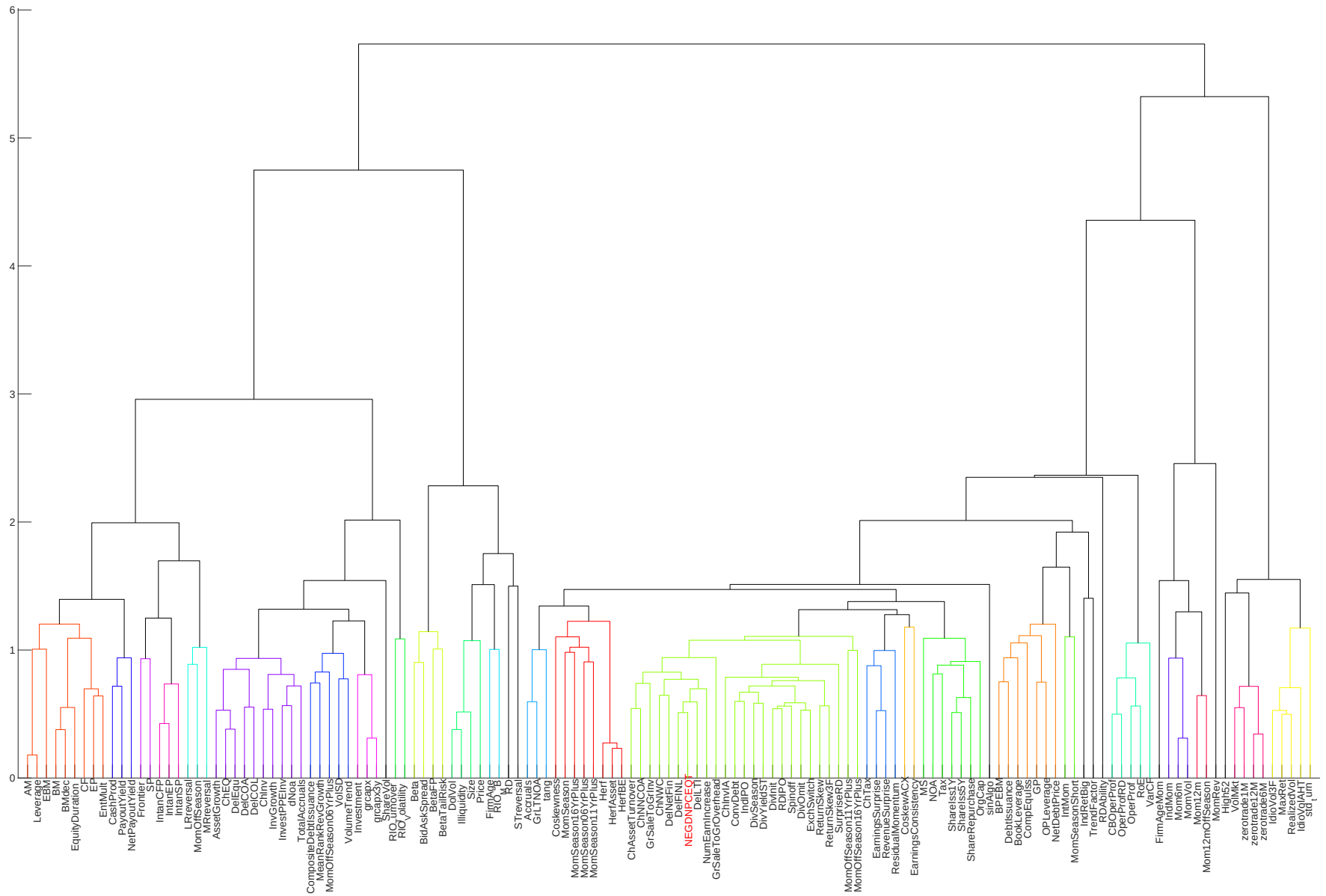


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

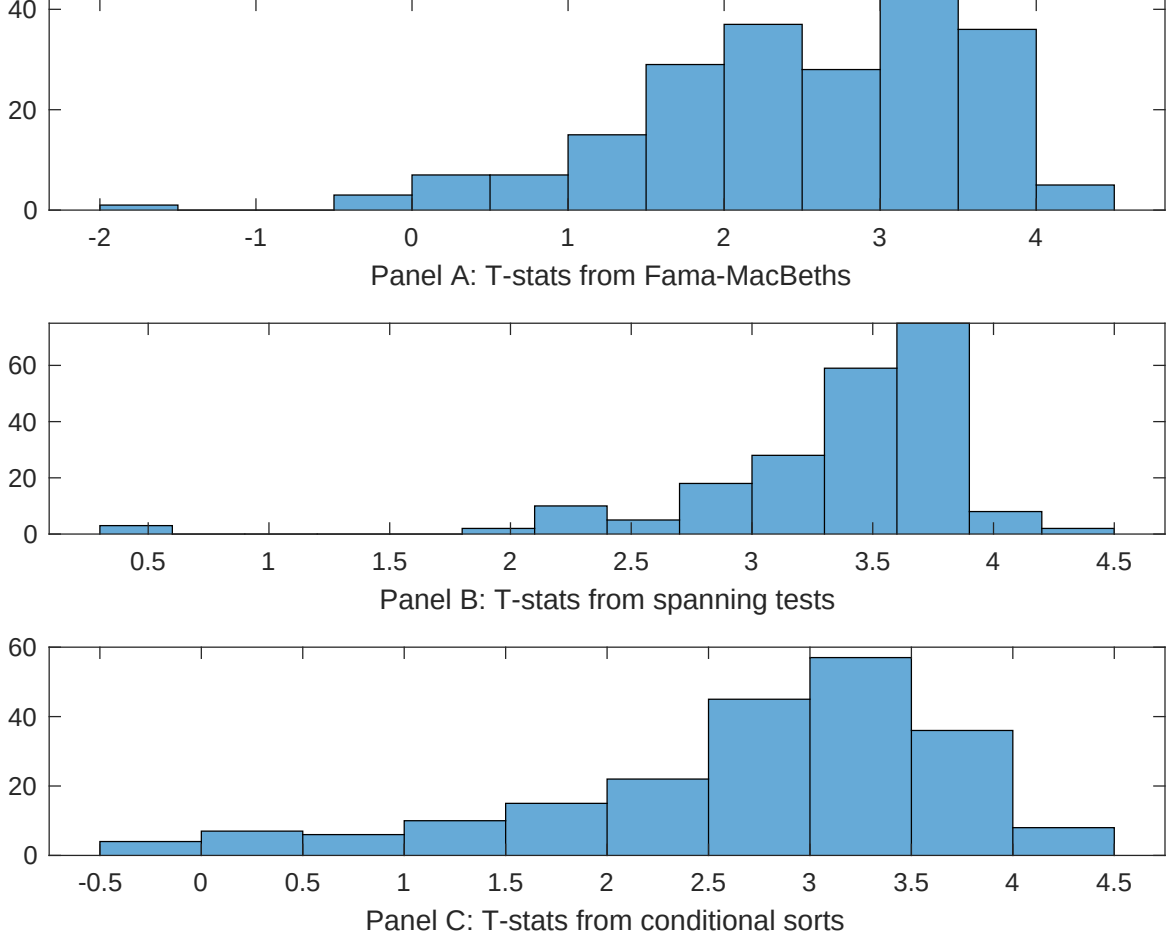


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of DCP conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{DCP} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{DCP} DCP_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{DCP,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on DCP. Stocks are finally grouped into five DCP portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted DCP trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on DCP. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{DCP}DCP_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Change in financial liabilities, Accruals, Change in Net Working Capital, Net debt financing, Investment to revenue, Change in Net Noncurrent Op Assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.12 [5.76]	0.11 [5.38]	0.12 [5.49]	0.12 [5.12]	0.15 [6.59]	0.12 [5.50]	0.14 [5.61]
DCP	0.69 [1.02]	0.15 [2.36]	0.20 [3.13]	0.14 [2.20]	0.13 [1.65]	0.22 [3.24]	-0.23 [-0.04]
Anomaly 1	0.17 [9.25]						0.59 [1.93]
Anomaly 2		0.13 [4.17]					0.21 [3.70]
Anomaly 3			0.62 [2.31]				-0.15 [-2.90]
Anomaly 4				0.21 [9.37]			0.90 [2.47]
Anomaly 5					0.24 [6.36]		0.15 [3.52]
Anomaly 6						0.75 [4.03]	0.16 [0.69]
# months	732	732	732	630	732	732	630
$\bar{R}^2(\%)$	0	0	0	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the DCP trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{DCP} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Change in financial liabilities, Accruals, Change in Net Working Capital, Net debt financing, Investment to revenue, Change in Net Noncurrent Op Assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.14 [2.42]	0.12 [2.02]	0.13 [2.24]	0.17 [2.79]	0.16 [2.63]	0.16 [2.77]	0.11 [2.00]
Anomaly 1	30.04 [9.09]						27.23 [6.18]
Anomaly 2		16.08 [6.80]					11.41 [4.61]
Anomaly 3			14.54 [4.92]				3.53 [1.12]
Anomaly 4				14.58 [4.28]			-8.49 [-2.04]
Anomaly 5					11.87 [4.95]		3.26 [1.39]
Anomaly 6						17.28 [6.46]	9.55 [3.37]
mkt	-1.72 [-1.27]	-1.20 [-0.86]	-2.60 [-1.84]	-1.01 [-0.70]	-2.63 [-1.87]	-2.47 [-1.77]	0.25 [0.18]
smb	4.67 [2.34]	11.02 [5.34]	8.33 [4.08]	7.25 [3.33]	5.39 [2.58]	6.90 [3.42]	7.46 [3.50]
hml	-8.32 [-3.15]	-7.40 [-2.71]	-9.60 [-3.51]	-9.80 [-3.56]	-9.81 [-3.60]	-11.30 [-4.19]	-5.71 [-2.19]
rmw	-7.50 [-2.82]	0.53 [0.19]	-3.43 [-1.24]	-5.61 [-1.98]	-3.95 [-1.43]	-7.00 [-2.56]	-2.31 [-0.83]
cma	10.59 [2.61]	17.35 [4.33]	21.33 [5.32]	18.18 [4.24]	20.80 [5.18]	21.94 [5.54]	10.40 [2.54]
umd	2.96 [2.15]	3.98 [2.86]	4.56 [3.24]	4.59 [3.18]	3.97 [2.79]	3.31 [2.33]	1.18 [0.86]
# months	732	732	732	630	732	732	630
$\bar{R}^2(\%)$	19	15	12	13	12	14	26

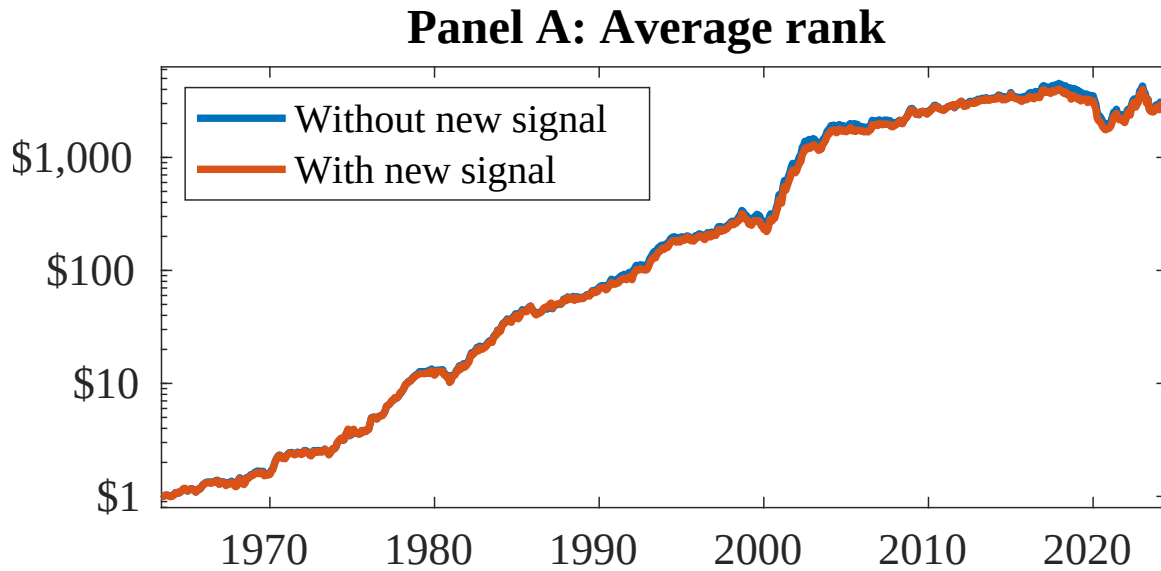


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as DCP. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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